

Answer: D

$$\mathbf{27} \quad R = \frac{\rho L}{A} = \frac{\rho 30 \pi d_{rod}}{\pi \left(\frac{d_{wire}}{2} \right)^2} = \frac{1.3 \times 10^{-8} (30) \pi (1.5 \times 10^{-2})}{\pi (0.25 \times 10^{-3})^2} = 9.4 \times 10^{-2} \Omega$$

Answer: B

$$\mathbf{28} \quad \text{Total resistance in circuit} = \frac{1}{\frac{1}{1} + \frac{1}{1}} + 3 = 6 \Omega$$